

RC Autonomous Car: Budget and Options

FEB Autonomous Team

Formula Electric at Berkeley

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The ask

One 1:10-scale autonomous car on the RoboRacer (F1TENTH) standard: **\$1,835 recommended**, \$1,105 minimum using sensors we already own, \$3,045 for the official reference build. The FSAE car is on a two-year cycle and cannot host autonomous work this season. The RC car is where the team develops, tests and learns in the meantime, and the software pipeline built on it is the one that goes onto the FSAE car next year.

1 What the car is and how it drives

Part	Role
Chassis	Traxxas Fiesta ST Rally VXL, ready to run: Slash 4x4 chassis, 4WD, brushless motor, steering servo. The chassis the current RoboRacer build guide and deck drawings are made for, and the car our simulator models.
Motor controller	VESC-type. Drives motor and servo from USB and reports wheel rotation back as speed.
Computer	Runs ROS 2: sensor drivers, perception, planning, control.
2D lidar	A range per angle, 5 to 40 times a second. Sees walls and cones; enough to drive with no map.
Camera	Colour: which cone is which. Needed for the perception work.
Joystick	Bluetooth gamepad for manual driving and the dead-man switch: the car drives autonomously only while a button is held.
Power	One 3S lithium pack drives the motor; the computer takes pack voltage through a fuse; a 5 V converter feeds lidar and camera.
Deck	Laser-cut plates and standoffs from the published drawings.

Flow: lidar, wheel odometry and camera in; perception locates walls or cones; a planner picks a bearing and speed; a controller turns them into steering and throttle; the VESC actuates, gated by the dead-man. Same nodes and topic names as the team's simulator, so a stack developed there drops onto the car with only the driver nodes swapped.

Budget

Item	Minimum	Recommended	Reference
Chassis: Traxxas Fiesta ST Rally VXL	430	430	430
Controller: Flipsky Mini FSESC 6.7 / VESC 6 MkVI	110	110	250
Compute: Raspberry Pi 5 8 GB kit / Jetson Orin Nano Super + 500 GB NVMe	130	350	350
Lidar: RPLidar A1M8 (owned) / RPLidar S2 / Hokuyo UST-10LX	0	400	1,200
Camera: Logitech C920 / Luxonis OAK-D Lite / RealSense D435i	60	150	335
Batteries: two Zeee 3S 6000 mAh, balance charger, LiPo bag	135	135	135
Power distribution: 5 V converter, barrel-jack pigtail, balance extender, fuse, switch / RoboRacer powerboard	40	40	125
Joystick, antenna, PPM cable, bullet adapters, USB cables, display emulator	70	70	70
Deck plates (cut on campus from the RoboRacer drawings), standoffs, fastener kits	90	90	90
Spares: tyres, pinion, bumper	40	60	60
Total	1,105	1,835	3,045

Prices are September 2026 list prices, rounded. **Minimum** drives slow laps on the RPLidar we own and a Pi. **Recommended** is the official build with a time-of-flight RPLidar and a Flipsky controller in place of the two expensive parts; every item is one the team keeps, and the lidar upgrades later by bolting on. **Reference** is the official RoboRacer competition bill of materials.

Options in each area

Lidar

The lidar sets the speed the car can drive: at 10 Hz it moves 25 cm between scans at 3 m/s. All publish the same ROS message, so a swap changes no code; the RPLidar sits on a small printed adapter plate and the Hokuyo bolts straight to the deck.

Lidar	USD	Rate	Range	Reality
RPLidar A1M8 (owned)	0	5 to 10 Hz	12 m	laps at 1 to 2 m/s; 1° resolution, coarse maps; runs the car until the S2 arrives
RPLidar A2M12	320	12 Hz	12 m	fine indoors; noisy on dark surfaces
RPLidar S2	400	10 Hz	30 m	time of flight; clean maps; to 3 m/s
RPLidar A3	600	15 to 20 Hz	25 m	fastest cheap unit
Hokuyo UST-10LX	1,200	40 Hz	10 m	the reference; what race winners run

Compute

Computer	USD	On the RC car	On the FSAE car
Raspberry Pi 5 , 8 GB	130	lidar-only driving; no camera perception	no
Jetson Orin Nano Super , 250 + 100 NVMe 8 GB		the official part; fits the deck; enough for everything the car does	no: a 3D lidar and several cameras need more
Jetson Orin NX 16 GB on the Nano carrier	about 900	fits; overkill	a first driverless stack: M1 lidar, two cameras, detector
Jetson AGX Xavier 16 GB (owned, 2018)	0	no: 630 g box, does not fit the deck	a bridge for one or two seasons; last software is JetPack 5
Jetson AGX Orin 64 GB	2,000	no: 1.4 kg	the full answer, when the stack needs it

Recommended: the Orin Nano on the RC car, permanently. The FSAE computer is decided next year with a season of data; the owned Xavier carries FSAE perception development until then. If the club prefers one purchase for both cars, the Orin NX is the largest that fits the RC car and moves to the FSAE car later, at about \$650 more, leaving the RC car needing a Nano then.

Why a Jetson from day one. Jetson to Jetson is the same software: same operating system image, same ARM64 build, same CUDA and TensorRT, same ROS 2 packages. Moving the stack from the Orin Nano to an Orin NX or AGX Orin is copying it across. Developing on an x86 laptop or PC instead means a different architecture, different driver and CUDA versions, and recompiling everything, with the porting landing at the worst moment, when the FSAE car is finally ready. Building on the Nano now makes the later upgrade trivial.

A separate line for the on-car FSAE computer. If the budget process allows it, the FSAE car's computer should be its own item, requested once the team has a season of data on what the stack needs: about \$900 for an Orin NX 16 GB or about \$2,000 for an AGX Orin. It is not part of this ask, and buying it now would mean guessing.

Camera

[Logitech C920](#), \$60, rolling shutter, fine at these speeds. [OAK-D Lite](#), \$150, stereo depth and on-camera inference, so cone detection costs the computer nothing; the recommended pick. [OAK-D Pro W](#) or [ZED 2i](#), \$400 to \$500, global shutter and weatherproof: FSAE-grade cameras that also fit the RC car, if the club wants the camera bought once.

Controller

[Flipsky Mini F5ESC 6.7](#), \$110: same firmware as the reference, what most student builds use. [VESC 6 MkVI](#), \$250: the reference part, better cooling and support. Either is a drop-in.

4 What we already own, and where it goes

- **RPLidar A1M8.** Onto the RC car for the first month: laps at 1 to 2 m/s until the S2 arrives, then a spare.
- **RoboSense M1 3D lidar.** It can be bolted to the RC car on a custom plate, and the car would drive, badly: 0.7 kg on a 3 kg car spoils the handling, its half-metre blind zone hides the

nearest wall in a 2 m corridor, and its 120° forward view gives wall-following nothing to work with. It is an FSAE sensor; with the Xavier it forms a perception rig for cone-track data this semester.

- **AGX Xavier 16 GB.** Computer for that rig and a bridge for the FSAE car; not for the RC car.
- **RealSense D435i.** Second camera on the rig.
- **Tamiya 1:10 chassis.** Motorless touring kit: no drawings, no match to the simulator, a semester of mechanical work to save \$300. Bench mule.
- **Alameda test track.** Outdoor sessions for the rig and, later, the FSAE car; not needed for the RC car, which runs in a corridor.

5 Why it is worth it

This semester

- The FSAE car is being built and cannot host autonomy; without a platform the team has nothing to run on until next year.
- Members stay for what they can see move.
- Real dynamics and real sensors cannot be simulated: grip, servo backlash, battery sag, a lidar that sees chair legs.
- New members learn wiring, power, safety, motor control and ROS 2 on hardware at low cost and low risk.
- The simulator and the car share one codebase, so every member's stack has somewhere to go.

Carrying over to the FSAE car

- **Same pipeline.** Cone detection with camera plus lidar, midline between cones, raceline, tracking controller: the Formula Student Driverless stack at small scale.
- **Same tools.** ROS 2 nodes and topics, sensor drivers, bag recording and replay, sim-to-real testing, safety procedures.
- **Same method.** Develop in the simulator, transfer to the car, measure the gap, close it.
- **What does not transfer:** vehicle dynamics and tuned parameters. A 3 kg car says nothing about a 250 kg car's tyres.

6 Will it work, and the practical realities

The build is the F1TENTH/RoboRacer platform, assembled by hundreds of university teams since 2016 from a public build guide; the parts list is that guide's own with cheaper sensor and controller choices. Driving laps is a solved problem, and our simulator already runs the same car and software. The car can run the whole pipeline: mapping, localisation, a raceline and a proper controller; the cheaper sensors cap speed, not capability. The recommended car is eligible for RoboRacer physical events as built; a lidar upgrade is what would make it fast.

- **One owner.** The car needs a named person for assembly, batteries and access, or it goes unused. This is the main risk.
- **Time.** Two to four weekends to assemble and integrate; first autonomous lap within a month; mapping, racelines and perception fill the semester.
- **Speed.** 1 to 2 m/s on the A1M8, about 3 m/s on the S2, faster only with the Hokuyo. The simulator car goes faster; members will feel the gap.
- **Access.** One car; time on it is earned through the simulator's qualification rule.
- **Batteries.** LiPo bag, supervised charging, a written procedure before the first charge.
- **Wear.** Bumpers and servo savers break; the spares line covers it.

- **Not in this budget.** A 1:5-scale car (Traxxas X-Maxx, Arrma Kraton 8S, Losi 5IVE-T: about \$3,500 even with the M1 and Alameda counted, and not runnable on campus), a second car, and the FSAE computer. All later decisions.

RoboRacer build guide and bill of materials: <https://f1tenth.readthedocs.io>

Team simulator and manuals: <https://github.com/Pranman1/feb-racing>